

EXP: 2

DATE:

21/07/2026

Double sideband suppressed carrier

[DSB-SC]

AIM:

To generate the double sideband suppressed carrier by using SCILAB.

APPARATUS REQUIRED:

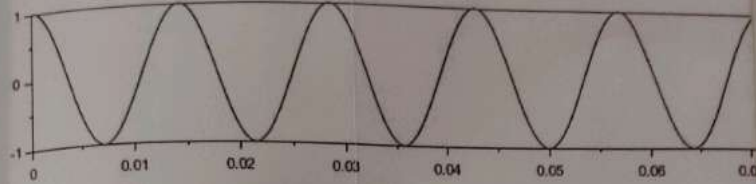
- * Computer with I3 processor
- * Scilab software.

CODE:

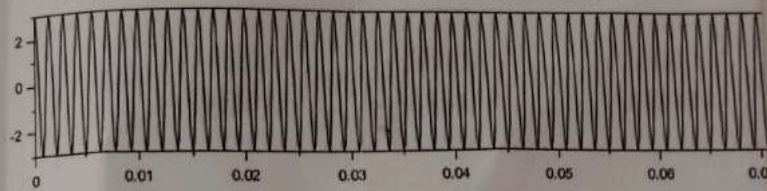
```
clc;  
clear;  
close;  
Am = 3;  
fm = 150;  
Ac = 6;  
fc = 1500;  
fs = 20000;  
t = 0:1/fs:0.04;  
m = Am * cos(2 * %pi * fc * t);  
subplot(3,1,1);  
plot(t, m);  
title("message signal");
```

S. No	Parameter	Theory		Practical	
		Amplitude	Frequency	Amplitude	Time Period
1	Message signal	3 V	150 Hz	3 V	$T_m = \frac{1}{f_m} = \frac{1}{150}$ $= 0.00667 \text{ s}$
2	Carrier signal	6 V	1500 Hz	6 V	$T_c = \frac{1}{f_c} = \frac{1}{1500}$ $= 0.000667 \text{ s}$
3	AM signal	-	-	18 V	$T_{Am} = \frac{1}{f_m} = \frac{1}{150}$ $= 0.00667 \text{ s}$

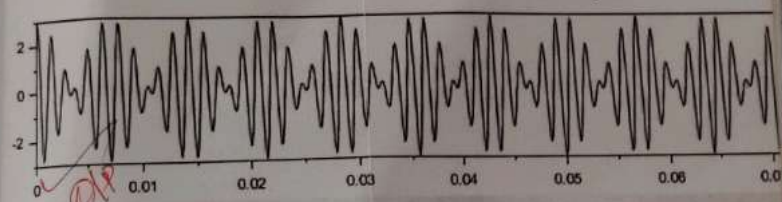
Message Signal



Carrier Signal



DSB-SC Signal



Calculation.

1. Message Signal:

Practical time

$$\begin{aligned}T_m &= x_2 - x_1 \Rightarrow (13.32 \times 10^{-2}) - (6.66 \times 10^{-2}) \\&= 6.66 \times 10^{-2} \text{ s} \\&= 0.0667 \text{ s}\end{aligned}$$

Practical Frequency:

$$\begin{aligned}f_m &= 1/T_m \\&= 1/0.0667 \\&= 150 \text{ Hz}\end{aligned}$$

2. Carrier Signal

Practical Time.

$$\begin{aligned}T_c &= x_2 - x_1 \Rightarrow (13.32 \times 10^{-3}) - (6.66 \times 10^{-3}) \\&= 6.66 \times 10^{-3} \text{ s} \\&= 0.0066 \text{ s}\end{aligned}$$

Practical frequency

$$\begin{aligned}f_c &= 1/T_c \\&= 1/0.0066 \\&= 1500 \text{ Hz}\end{aligned}$$

$c = A_c * \cos(2 * \pi * f_c * t);$

subplot (3, 1, 2);

plot (t, c);

title ("carrier signal");

dstb = m .* c;

subplot (3, 1, 3);

plot (t, dstb);

title ("DSB-SC signal");

Marks allocation

Scilab program implementation (20)	wave form and tabulation (20)	Calculation and result (20)	voice - (20)	Record observation (20)	Total marks
20	20	20	17	20	97

RESULT :

Thus, the Double sideband suppressed carrier (DSB-SC) signal was successfully generated using SCILAB.

3. DSB - SC signal

Practical Amplitude

$$\begin{aligned}\text{Peak Amplitude} &= A_m \times A_c \\ &= 3 \times 6 \\ &= 18 \text{ V}\end{aligned}$$

Practical Time,

$$\begin{aligned}T_{\text{DSB}} &= x_2 - x_1 \Rightarrow (13.32 \times 10^{-3}) - (6.66 \times 10^{-3}) \\ &= 6.66 \times 10^{-3} \text{ s} \\ &= 0.0066 \text{ s}\end{aligned}$$

Practical frequency

$$\begin{aligned}f_{\text{DSB}} &= 1/T_{\text{DSB}} \\ &= 1/0.0066 \text{ s} \\ &= 1500 \text{ s}.\end{aligned}$$